

1. Administrative & Study Identification

Full Study Title: Phase Ib/II Trial of Pembrolizumab (MK-3475) Combination Therapies in Metastatic Castration-Resistant Prostate Cancer (mCRPC) (KEYNOTE-365) **Short Title/Acronym:** KEYNOTE-365 **Protocol Number:** MK-3475-365 **NCT Number:** NCT02861573 **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Pembrolizumab (MK-3475, KEYTRUDA®) in combination with various agents (e.g., olaparib, docetaxel, enzalutamide, abiraterone, lenvatinib, vibostolimab [MK-7684A], carboplatin/etoposide [Etopophos, Vepesid, Toposar], belzutifan, prednisone) **Phase of Development:** Phase 1b/2 (PHASE1, PHASE2) **Medical Monitor:** MSD Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety and tolerability of pembrolizumab combination therapies (including the number of participants with adverse events [AEs] and discontinuing study drug due to AEs), and to evaluate the Prostate-Specific Antigen (PSA) response rate (percentage of participants with a $\geq 50\%$ decline from baseline) and Objective Response Rate (ORR) based on RECIST 1.1 per Blinded Independent Central Review (BICR). **Secondary Objectives:** To evaluate Radiographic Progression-Free Survival (rPFS), Overall Survival (OS), Time to PSA progression, Disease Control Rate (DCR) based on PCWG3-modified RECIST 1.1 assessed by BICR, and Duration of Response (DOR) based on PCWG3-modified RECIST 1.1 assessed by BICR.

2.2 Background & Rationale To address the gap in effective therapeutic options for patients with metastatic castration-resistant prostate cancer (mCRPC) whose disease has progressed despite treatment with next-generation hormonal agents or chemotherapy. Pembrolizumab has shown activity in mCRPC, and combining it with PARP inhibitors, targeted therapies, and chemotherapy may provide additional survival and clinical benefits by downregulating oncogenic pathways and enhancing anti-tumor immune responses.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** Recruiting **Control Type:** Uncontrolled and Controlled (depending on cohort; primarily single-arm or multi-arm parallel cohort assignments) **Blinding:** Open-label (Masking not used)

Randomization: Parallel Assignment (patients are assigned to specific cohorts based on prior therapies; some sub-cohorts are randomized)

3.2 Planned Enrollment, Cohorts & Duration Target \$N\$:

Approximately 1,200 participants **Number of Sites:** Multicenter, global clinical trial **Estimated Study Start:** November 17, 2016 **Estimated Primary Completion:** October 22, 2027 **Cohort Definitions & Amendment Status:** * **Cohort A:** Pembrolizumab + olaparib (Closed via Amendment 08) * **Cohort B:** Pembrolizumab + docetaxel + prednisone (Closed via Amendment 08) * **Cohort C:** Pembrolizumab + enzalutamide (Closed via Amendment 08) * **Cohort D:** Pembrolizumab + abiraterone + prednisone (Closed via Amendment 08) * **Cohort E & F:** Pembrolizumab + lenvatinib (Closed via Amendment 14) * **Cohort G & H:** Pembrolizumab/vibostolimab coformulation (MK-7684A) (Closed via Amendment 14) * **Cohort I:** Pembrolizumab + carboplatin + etoposide (Arm 1) and carboplatin + etoposide (Arm 2) * **Cohort J:** Belzutifan (Arm 1) and Pembrolizumab + belzutifan (Arm 2) (Closed via Amendment 16)

(Note: Closure of cohorts A-H and J was not due to safety issues. No further efficacy and survival follow-up assessments will be collected in Cohorts A-H).

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male participants ≥ 18 years old with histologically or cytologically confirmed metastatic adenocarcinoma of the prostate without small cell histology (for Cohorts A, B, C, D, E, G, J). For Cohorts F, H, I: Has t-NE or de novo metastatic prostate cancer defined by $\geq 1\%$ neuroendocrine cells confirmed by central histology. 2. Confirmed disease progression within 6 months prior to screening (PSA progression: ≥ 2 rising PSA levels with ≥ 1 week interval, screening PSA ≥ 2 ng/mL; or radiographic progression in soft tissue/bone [≥ 2 new bone lesions]) while maintaining castrate levels of testosterone (<50 ng/dL) with ongoing androgen deprivation therapy (ADT). 3. Must be able to provide tumor tissue (core/excisional biopsy from soft tissue or bone, or archival tissue depending on specific cohort requirements). 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0, 1, or 2 for Cohorts A and C; performance status of 0 or 1 for Cohorts B, D, E, F, G, H, I, and J. 5. Must agree to use required contraception during the intervention period and for a specified time after the last dose (e.g., up to 95 days depending on the agent).

4.2 Exclusion Criteria 1. Known history of Human Immunodeficiency Virus (HIV), active Hepatitis B or Hepatitis C, or active autoimmune disease requiring systemic treatment in the past 2 years. 2. A "superscan" bone scan defined as intense symmetric activity in the bones and diminished renal parenchymal activity.

3. Prior treatment with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent, or prior participation in any other pembrolizumab trial.

4. Symptomatic congestive heart failure (NYHA Class III or IV), clinically significant pleural effusion/ascites, or a history of (noninfectious) pneumonitis requiring steroids. Cohorts E and F: Left ventricular ejection fraction (LVEF) below normal range or QTcF interval >480 milliseconds.

5. Prior solid organ or bone marrow transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab 200 mg administered via Intravenous (IV) infusion every 3 weeks (Q3W) on Day 1 of each cycle. Co-administered medications (e.g., olaparib, docetaxel + prednisone, enzalutamide, abiraterone + prednisone, lenvatinib, MK-7684A, carboplatin/etoposide, belzutifan) vary strictly by designated cohort. **Route of Administration:** Intravenous (pembrolizumab, docetaxel, carboplatin, etoposide, MK-7684A) and Oral (enzalutamide, abiraterone, olaparib, belzutifan, lenvatinib, prednisone). **Duration of Treatment:** Up to 35 cycles (approximately 2 years) for pembrolizumab/vibostolimab, or until documented disease progression, unacceptable toxicity, or withdrawal of consent. **Retreatment:** Participants who discontinue after 35 infusions (for reasons other than progression or intolerability) or after attaining a complete response may be eligible for a second course of treatment comprising up to 17 additional infusions (approximately 1 year) upon radiographic disease progression.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care, bone resorptive therapies (bisphosphonates/denosumab) if on stable doses for ≥ 4 weeks, and ongoing ADT. **Prohibited:** Live virus or live-attenuated vaccines within 30 days of study start, concurrent investigational agents, and strong CYP3A4 inhibitors/inducers for specific targeted therapy cohorts (e.g., Cohorts A, B, D).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue sample, Baseline Imaging, PSA Labs
Baseline	Day 1 (Cycle 1)	Cohort Assignment, First Dose of Study Treatments
Interim	Every 3 Weeks (Q3W)	Ongoing Treatment Dosing, Safety Labs, AE Monitoring
Imaging	Every 9 Weeks (Q9W)	CT/MRI and bone scans evaluated Q9W through Week 54, then Q12W thereafter
End of Study	Up to 35 Cycles	Final treatment, 30-day and 90-day Safety Follow-up, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints

- 1. Safety and Tolerability:** Number of participants with Adverse Events (AEs) and number of participants discontinuing study drug due to AEs.
- 2. PSA Response Rate:** Percentage of participants with a $\geq 50\%$ decrease in Prostate-Specific Antigen (PSA) from baseline.
- 3. Objective Response Rate (ORR):** Assessed via RECIST v1.1 by Blinded Independent Central Review (BICR).

7.2 Secondary & Exploratory Endpoints

- 1. Disease Control Rate (DCR):** Based on Prostate Cancer Working Group 3 (PCWG3)-modified RECIST 1.1 assessed by BICR.
- 2. Duration of Response (DOR):** Based on PCWG3-modified RECIST 1.1 assessed by BICR.
- 3. Overall Survival (OS) and Radiographic Progression-Free Survival (rPFS)** per PCWG3-modified RECIST 1.1.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Evaluated systematically by investigators using CTCAE v4.0 throughout the study and for 30 days post-treatment. **Serious Adverse Events (SAEs):** Monitored up to 90 days after the last dose of study treatment, or 30 days if new anticancer therapy is initiated. **Data Monitoring Committee (DMC):** Evaluates cohort safety and efficacy thresholds internally/externally as per MSD sponsor protocols.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy and safety endpoints are evaluated in all allocated participants who received ≥ 1 dose of study treatment (As-Treated Population). **Sample Size Justification:** A multi-cohort adaptive design targeting up to ~1,200 subjects. Sample sizes per arm are powered to adequately estimate response rates (ORR, PSA) with precision. **Handling of Missing Data:** Standard survival censoring rules applied for time-to-event outcomes (OS, rPFS). Non-evaluable patients typically imputed as non-responders for ORR/PSA endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite and remote clinical monitoring of protocol adherence and source data verification. **Audit Trail:** Electronic Case Report Form (eCRF) locking mechanisms, real-time data clarification tracking, and Blinded Independent Central Review (BICR) for imaging assessments.

11. Study Identifiers & Terminology

Primary ID: MK-3475-365 **Registry Number (NCT):** NCT02861573
Posting ID: 19839414555992495 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** KEYNOTE-365 **Official Regulatory Title:** Phase Ib/II Trial of Pembrolizumab (MK-3475) Combination Therapies in Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Condition Coding:** UMLS: Castration-Resistant Prostate Cancer | MeSH: D000085434 | SNOMED CT: 789498007 **Medication Classification:** Etoposide ATC Code: L01CB01

12. Clinical Framework (Prostate Cancer Specific)

Primary Condition: Metastatic Castration-Resistant Prostate Cancer (mCRPC) **Diagnosis Threshold:** Radiographic progression or PSA progression with castrate levels of testosterone (ongoing ADT) **Medical Methodology:** Immunotherapy combined with Chemotherapy, PARP Inhibition, Targeted Therapy, or Next-Generation Hormonal Agents **Optimization Target:** Objective response rate (ORR) and $\geq 50\%$ decline in PSA

13. Study Procedures & Methodology

Management Pathway: Multi-cohort advanced oncology clinical pathway for heavily pre-treated mCRPC **Education Protocol (HCP):** Trial-specific clinical initiation visits and safety monitoring training **Education Protocol (Patient):** Informed consent outlining specific treatment cohort risks/benefits and immune-mediated AE awareness **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for radiological assessments

14. Observation & Follow-Up Schedule

Total Duration: Up to 35 treatment cycles (~2 years) plus potential retreatment phase for an additional 17 cycles (~1 year) + ongoing survival follow-up **Onsite Visit Cadence:** Every 3 weeks (Q3W) matching dosing cycles **Remote Monitoring Frequency:** Intermittent survival and post-treatment follow-up calls **Checklist Review Interval:** Scans every 9 weeks (Q9W) to week 54, then every 12 weeks (Q12W)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				